BOARD / CXO BRIEFING

Predictive + Prescriptive Maintenance

Convert OT telemetry into bottom-line impact: fewer failures, faster decisions, and measurable EBITDA protection across industrial operations.

Business Goal

- Reduce unplanned downtime and production loss
- Prioritize high-value interventions first
- Align operations and finance on one decision view
- Equip AEs and SAs with one repeatable demo-to-value motion

Databricks Outcome

- Unified OT+IT+financial data foundation
- Real-time risk and RUL intelligence
- Action recommendations with economic impact
- Same solution pattern across 5 industries: Mining, Energy, Water, Automotive, Semiconductor

WHY THIS MATTERS

Predictive Alone Tells You What. Prescriptive Tells You What to Do.

Without Prescriptive

- Too many alerts, low actionability
- Engineering and finance misaligned
- Interventions triggered too early or too late

With Predictive + Prescriptive

- Risk scored by failure likelihood and timing
- Actions ranked by value at risk
- Decision captured: approve, defer, or reject

30-50%
FEWER CRITICAL SURPRISES

10-20%
LOWER MAINTENANCE SPEND WASTE

2-4x
FASTER TRIAGE DECISIONS

High
EXECUTIVE CONFIDENCE

BOTTOM-LINE VISIBILITY

Financially-Quantified Maintenance Decisions

Every recommendation is tied to economic outcome: avoided loss vs intervention cost, in business currency.

Signal drift -> Failure risk + RUL -> Prescribed action -> Avoided loss projection -> Recorded operator decision

Example Asset Decision

- Avoided cost: JPY 1,849,543
- Intervention cost: JPY 507,133
- Net protected value: JPY 1,342,410

Executive Readout

- Portfolio risk concentration
- Expected loss if deferred
- Actioned vs unactioned exposure

DATABRICKS END-TO-END

How the Platform Makes This Possible

OT Sources (simulator / Zerobus / PI) -> Delta Bronze -> Delta Live Tables (Silver/Gold) -> MLflow models (anomaly + RUL) -> FastAPI + React app -> AI-assisted actions (Genie)

Data + AI Stack

- Streaming and batch on one lakehouse
- Feature pipelines with lineage
- Model lifecycle, scoring, and traceability

Operational Decision Layer

- Geo and fleet risk visualization
- Asset-level recommendations and chat
- Action persistence for audit and execution
- Currency switch drives localized narrative and financial context (for example, JPY)

Field Readiness for AE + SA

- One end-to-end architecture and talk track reusable across Mining to Semiconductor
- Localization switch (currency/language) makes value story customer-native in every region

OT Sources
Zerobus / PI / Simulator
Delta Bronze
Raw + normalized ingest
DLT Silver / Gold
Features + predictions + value
ML + MLOps
MLflow anomaly + RUL
App + Genie
Geo, Fleet, AI actions
Business
EBITDA protection
Unity Catalog Governance
Lineage + Quality
Action Audit Trail
Currency/Language Localization
Lakehouse + AI
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GOVERNANCE AND TRUST

Enterprise Controls Built In

Data Governance

- Unity Catalog for access and policy control
- Lineage across OT tables, features, and outputs
- Consistent schemas across industries

AI Governance

- Versioned models and reproducible scoring
- Action-level audit trail (approve/defer/reject)
- Human-in-the-loop decision authority

Security | Compliance | Observability | Traceability

Governed AI is what makes maintenance recommendations board-ready, regulator-ready, and operations-ready.

5-MINUTE DEMO STORY

What You Show a CXO Live

- **1.** Open Geo view -> select a high-risk site and asset.
- **2.** Show anomaly + RUL + financial impact in selected currency.
- **3.** Ask AI for root cause, risk level, and action sequence.
- **4.** Execute decision (approve/defer/reject) and capture trace.
- **5.** Return to fleet view to show portfolio-level business impact.

Talk track cue: "This is not a dashboard. This is a governed, financially-aware decision system."

BUSINESS KPI FRAMEWORK

How Success Is Measured in 30-90 Days

-X%
UNPLANNED DOWNTIME HOURS

+Y%
PLANNED INTERVENTION RATIO

+Z
AVOIDED-LOSS RUN RATE

T-
DECISION LEAD TIME

Operational KPIs

- Critical alert closure time
- Mean time between disruptions
- Maintenance schedule adherence

Financial KPIs

- Avoided cost vs intervention spend
- Loss prevented per site / line / tool
- Quarterly EBITDA protection signal

EXECUTIVE CLOSE

Decision Request

Approve a focused pilot for one critical site and asset class, with baseline and target KPIs tracked weekly.

Pilot Scope

- 1 site, 1 asset class, 30-45 days
- Unified OT + maintenance + finance view
- Weekly governance review and KPI report

Expected Executive Output

- Documented value-at-risk reduction
- Productionized governance pattern
- Scale blueprint across plants/fabs

Bottom line: predictive tells you what may fail; prescriptive tells you what to do now; Databricks makes both measurable, governed, and scalable.